Swastya Setu — Project File Preview & Open Guide

Prepared from the current enclosure concept, supplied ESP32 firmware text, and firmware-derived pin map. This is a project draft package, not a validated vehicle-safety design.

Included project files

File type	Included file	Purpose / status
Mechanical source	.scad	Editable OpenSCAD concept enclosure
CAD exchange	.step (tray + lid)	Open in FreeCAD or import into SolidWorks
Mesh exports	.stl (tray + lid)	Easy viewing / 3D printing workflow
Electronics	.kicad_sch	Graphical system-level draft; not ERC-checked
Simulation	.m	Synthetic accident-detection and alert-state demonstration
Firmware reference	.txt	Copy of the firmware text supplied by the project owner

Why a .SLDPRT is not supplied

A genuine native SolidWorks part cannot be verified here. Open the STEP file in SolidWorks and use Save As → SLDPRT if you have SolidWorks installed. This produces an imported solid and does not recreate the original parametric feature history.

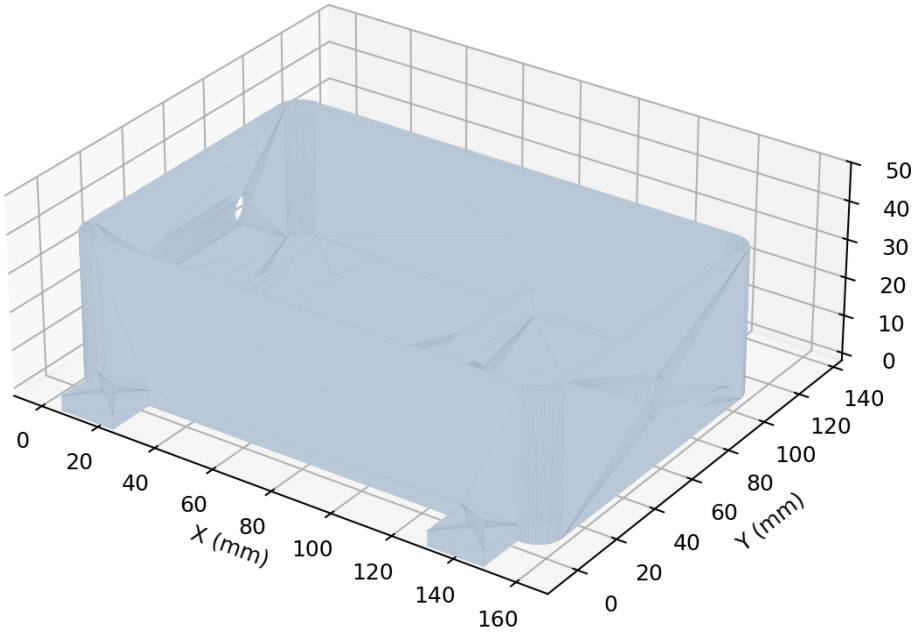
How to view files on your laptop

Want to see output now	Open the PNG previews in the preview folder, or view this PDF.
STEP / STL	Install FreeCAD (STEP) or use Windows 3D Viewer / Blender / a slicer (STL).
SCAD	Install OpenSCAD (free), open the .scad source, press F5 for preview or F6 to render.
KiCad	Install KiCad and open the .kicad_sch draft. Actual pins and electrical rules still need review.
MATLAB	Open the .m in MATLAB and run SwastyaSetu_AccidentAlert_Simulation. Thresholds are demonstration values.

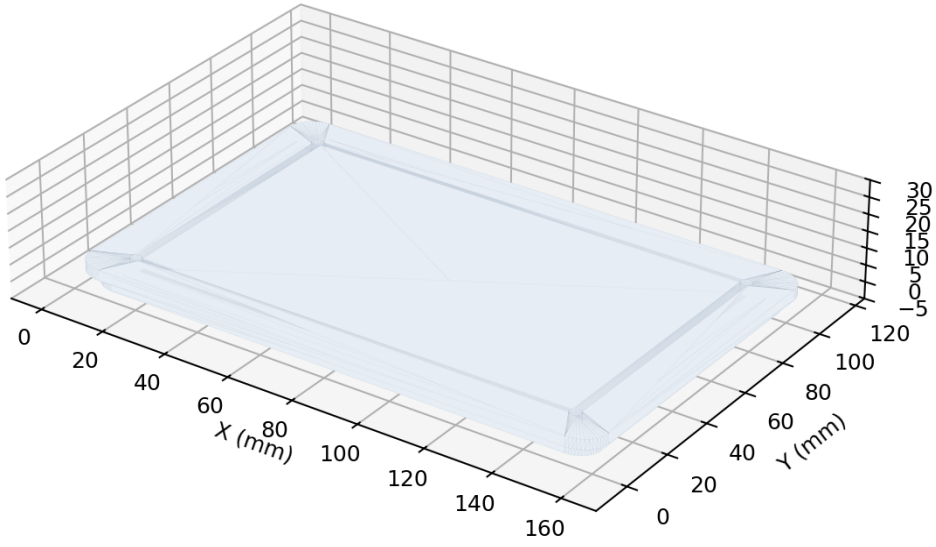
1. Enclosure preview

Swastya Setu — preliminary enclosure CAD preview

Enclosure tray (concept)



Enclosure lid (separate part)

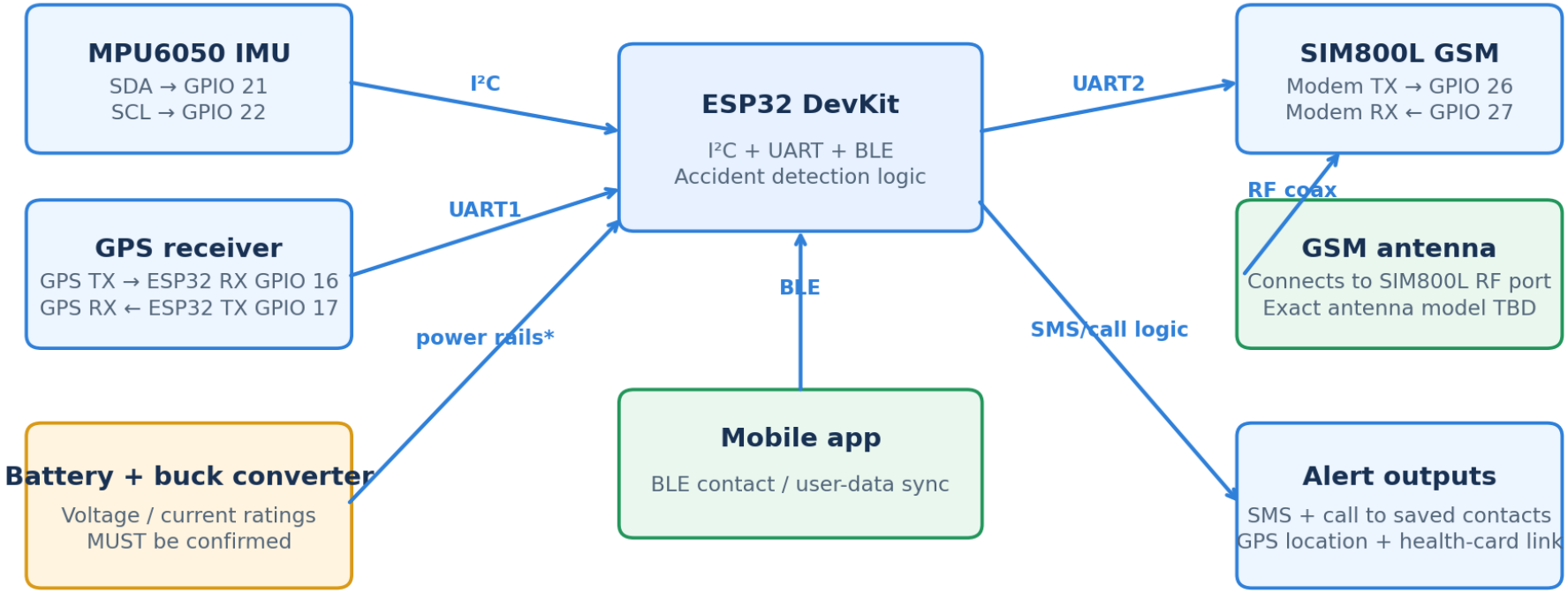


Concept geometry only. Check actual board/battery dimensions and fit before fabrication.

2. System connection preview

Swastya Setu — firmware-derived system connection preview

Draft pin map from supplied ESP32 firmware; physical wiring and power rails not yet verified

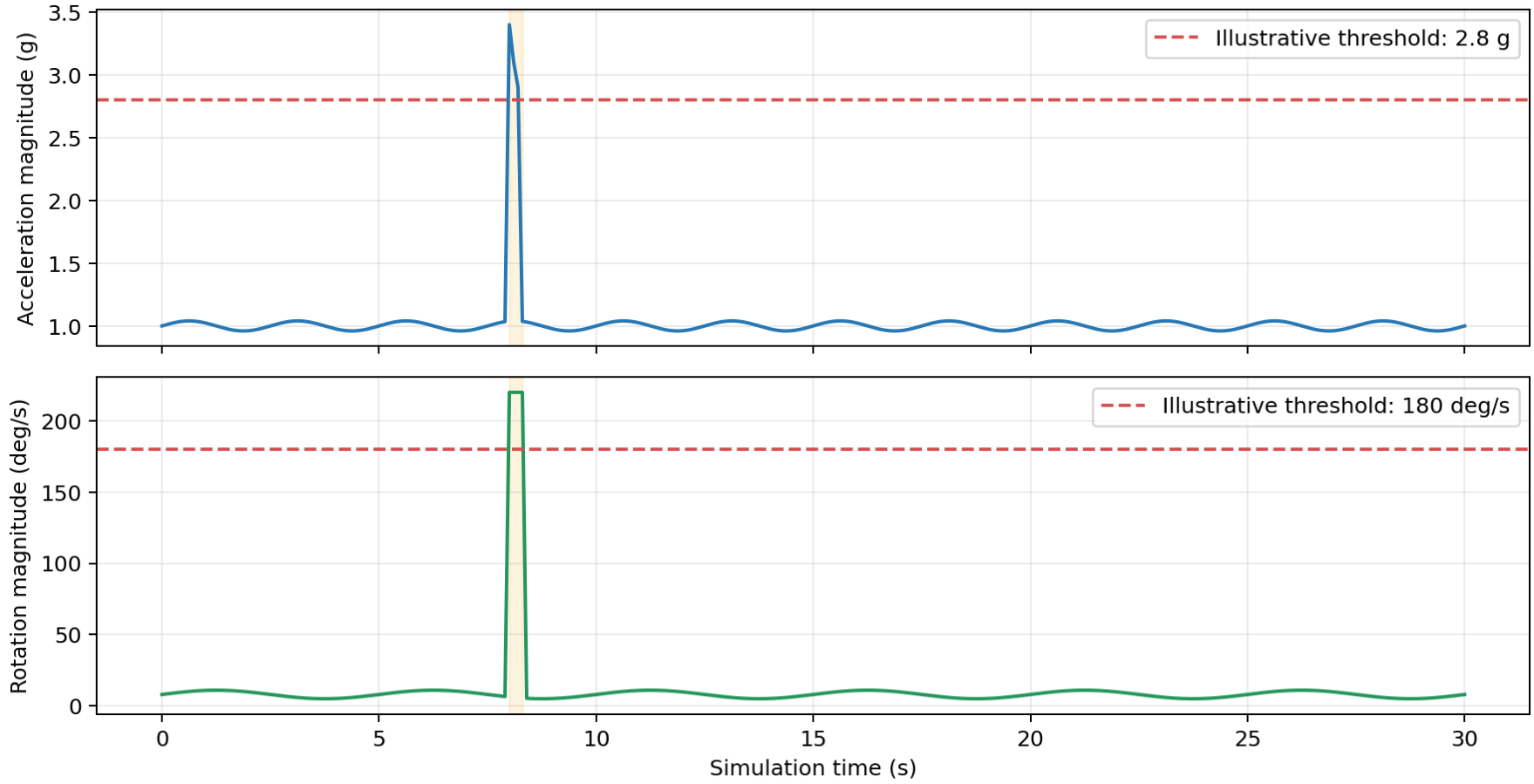


*Power distribution is conceptual only. Verify module voltage, SIM800L peak current, grounds and logic levels before powering.

Pin assignments are taken from supplied firmware. Power rails, grounds, actual board pin labels and physical connections are not yet verified.

3. Simulation output preview

Swastya Setu — illustrative MATLAB simulation output



Synthetic data only; not real vehicle testing and not validated accident-detection thresholds.

This is a visual recreation of the synthetic input data used in the MATLAB script, not an actual MATLAB execution log or measured vehicle data.

Validation and safety status

- STEP geometry imports as one solid per file; bounding boxes: tray approximately 160 × 134 × 38 mm, lid approximately 160 × 112 × 7 mm.
- No native SolidWorks .SLDPRT is supplied because a genuine native part could not be created and verified here.
- KiCad file is a graphical block-level draft. A KiCad ERC run and pin-accurate schematic review remain outstanding.
- MATLAB thresholds are illustrative only. The simulation does not establish real-world accident-detection accuracy.
- Firmware issue identified: lastLat and lastLon are declared but not updated before the SMS location URL is assembled.
- Do not mount this concept on or interfere with airbag/SRS sensors, modules, or wiring. Vehicle installation requires qualified safety review.

This package is intended to document the prototype concept and support continued engineering work. Do not describe unverified designs as tested or safety-certified.